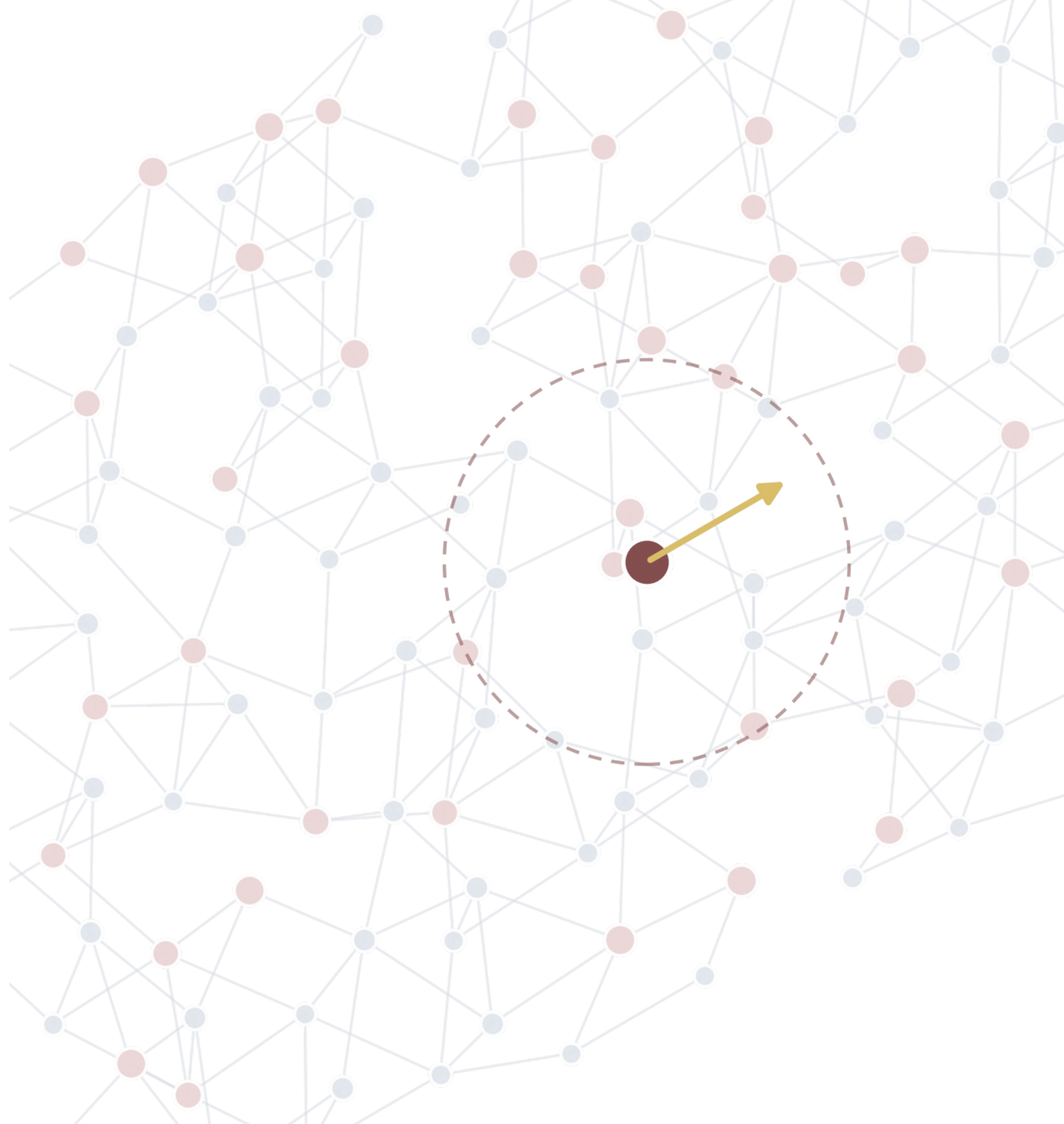


AGENTIC DISCOVERY & DEVELOPMENT · TEXAS A&M

MACE

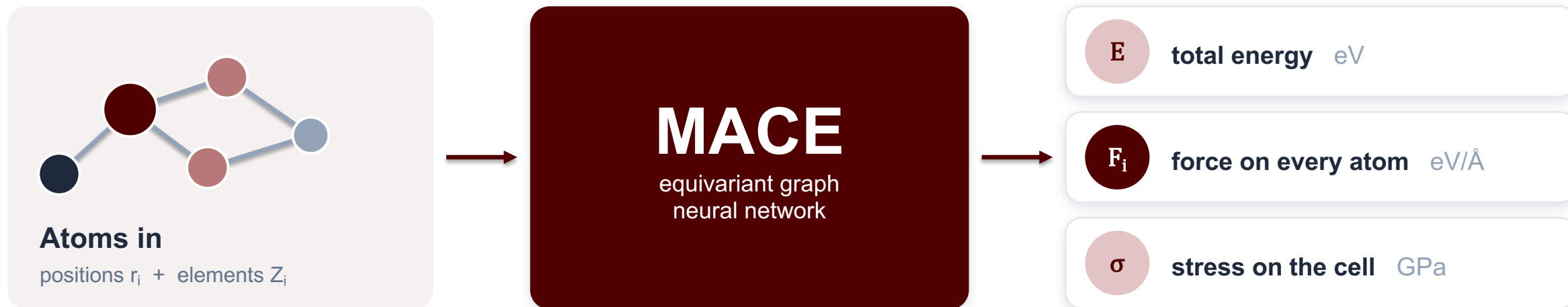
Machine-learned atomic forces as
an agent's physics engine

Joshua Jang · Fall 2026

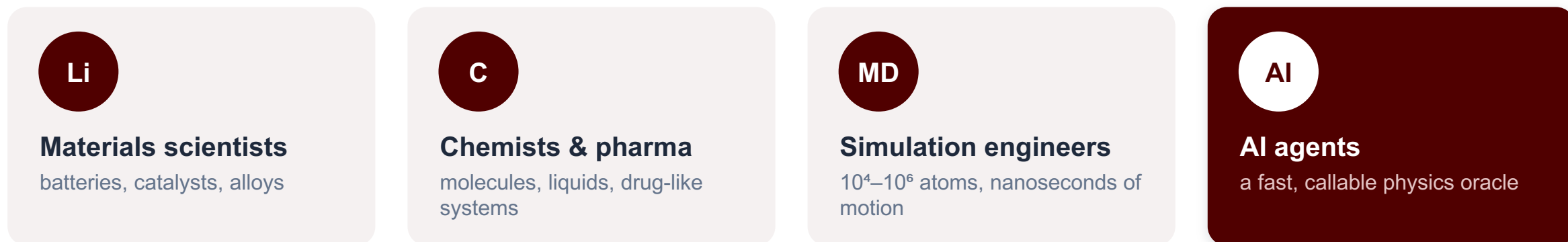


What is MACE?

A machine-learned interatomic potential: quantum-level accuracy at force-field speed

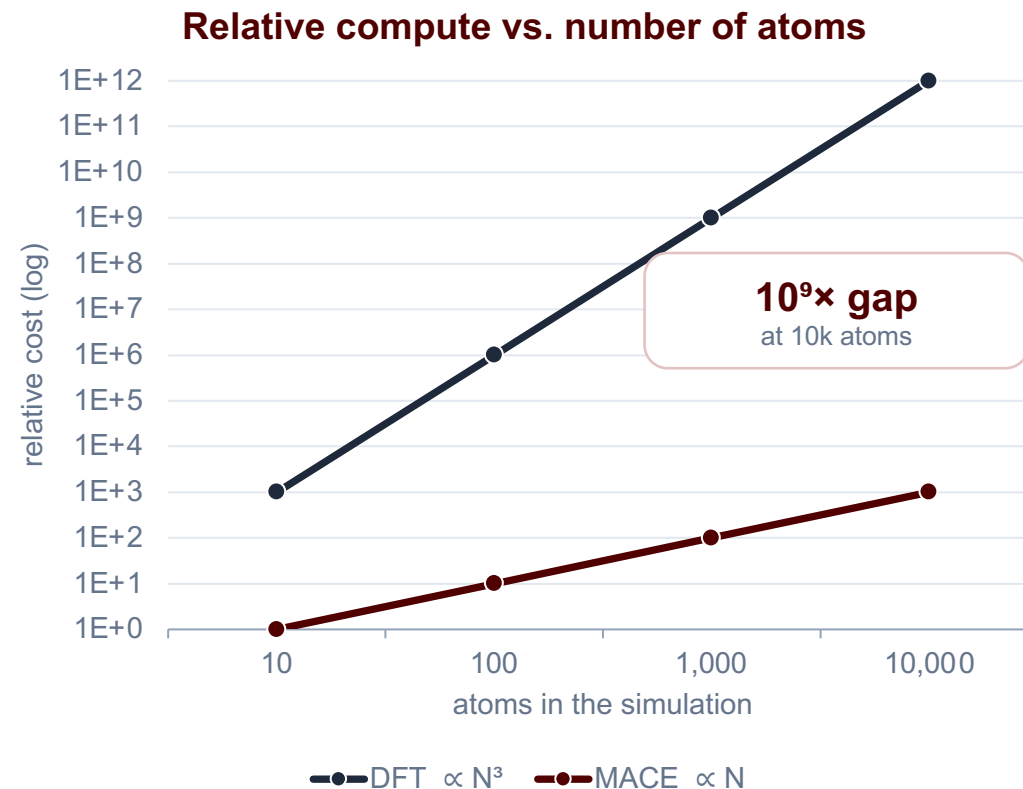
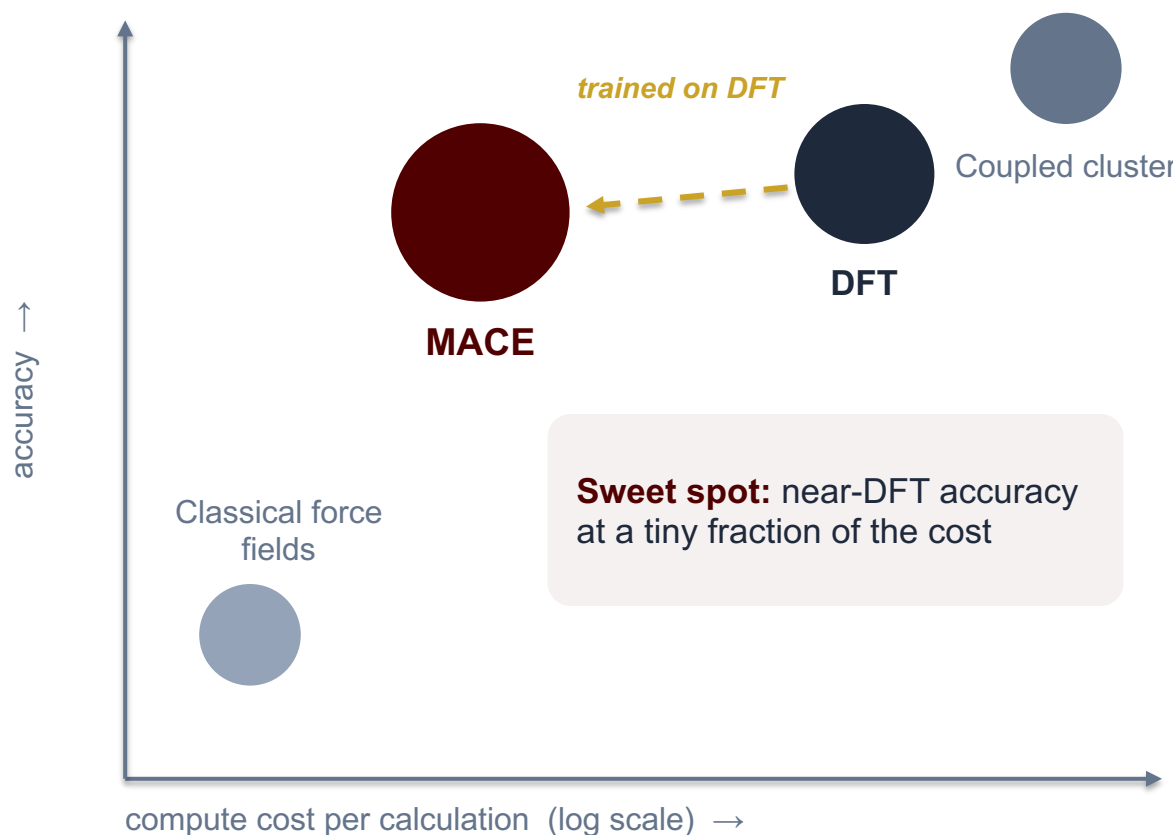


WHO USES IT



Where MACE sits: accuracy vs. cost

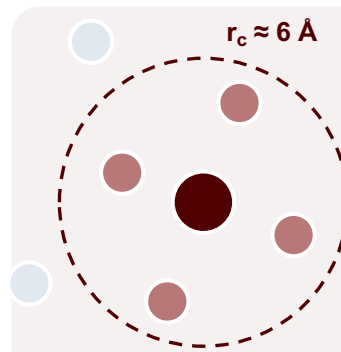
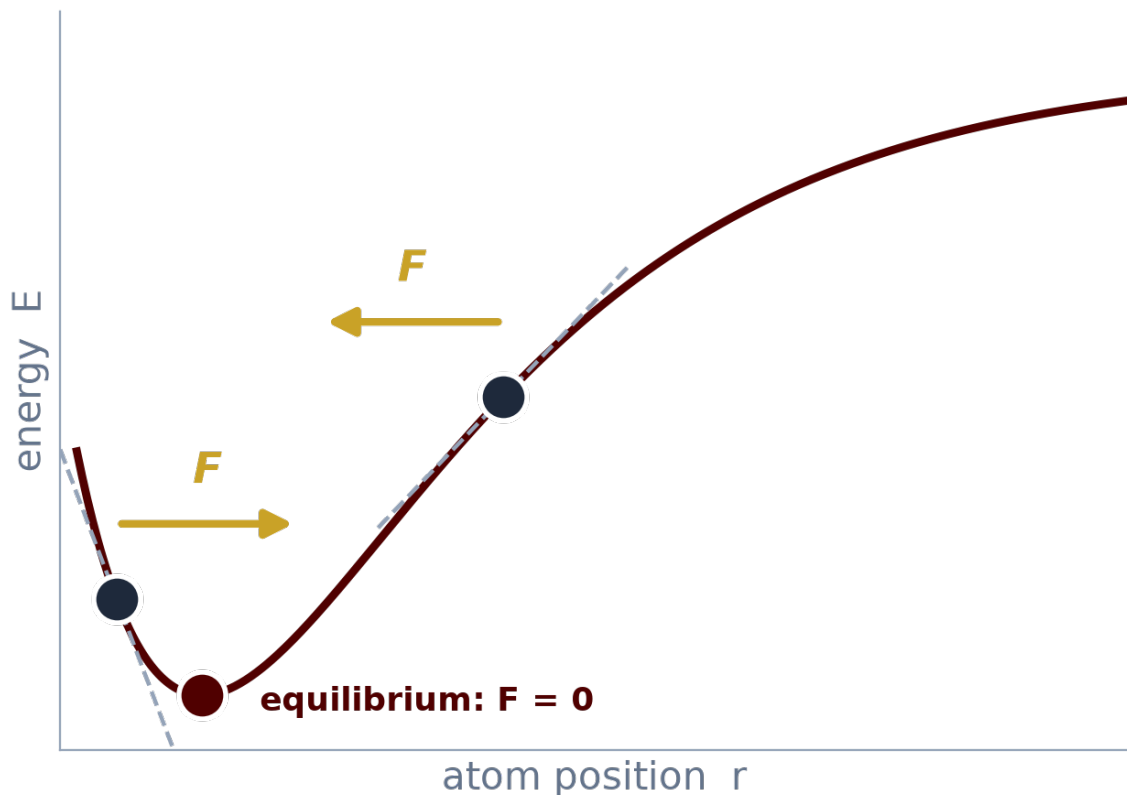
Learn from expensive quantum calculations once, then predict cheaply forever



MACE scales linearly: big systems and long simulations become possible

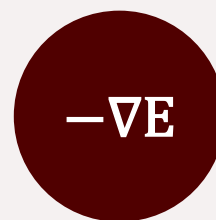
Forces are the slope of the energy

Learn one function, the energy, and every force follows by differentiation



$$E = \sum_i E_i(\text{neighbors within } r_c)$$

Total energy is a sum of local atomic contributions, so cost grows linearly



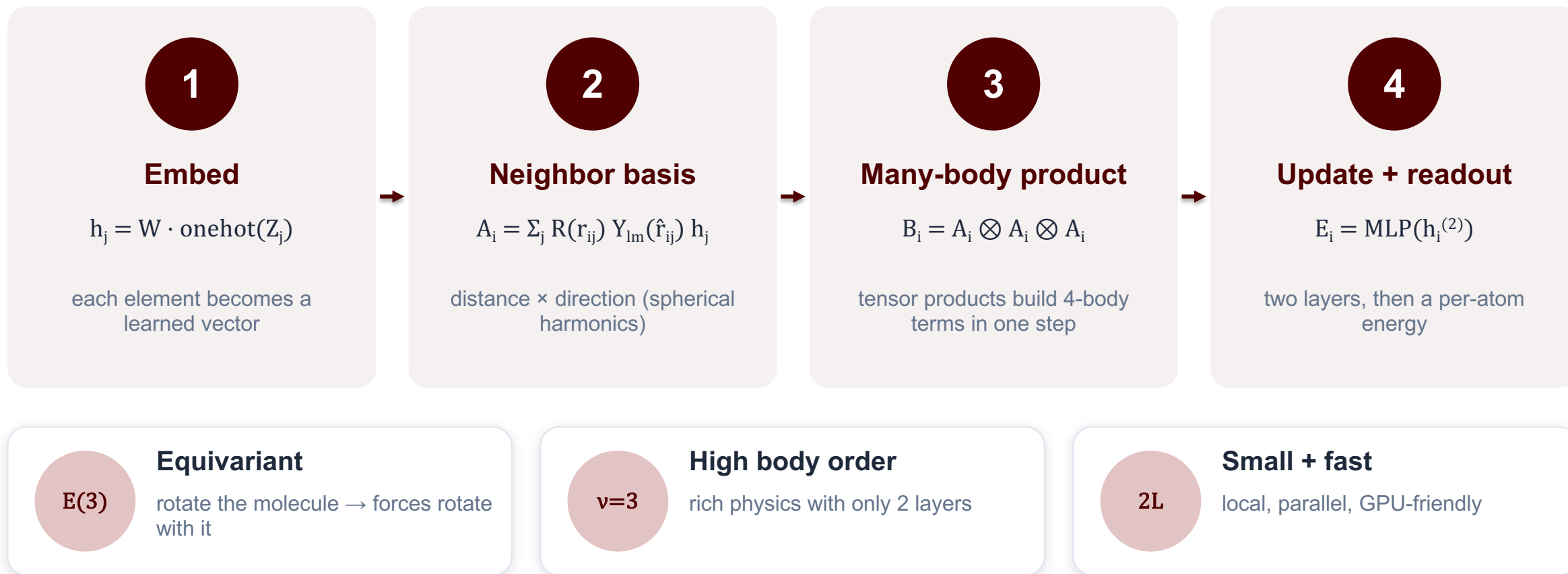
$$F_i = -\partial E / \partial r_i$$

Forces by automatic differentiation: energy is conserved, so dynamics stay stable

Train on DFT energies + forces → predict both for brand-new structures in milliseconds

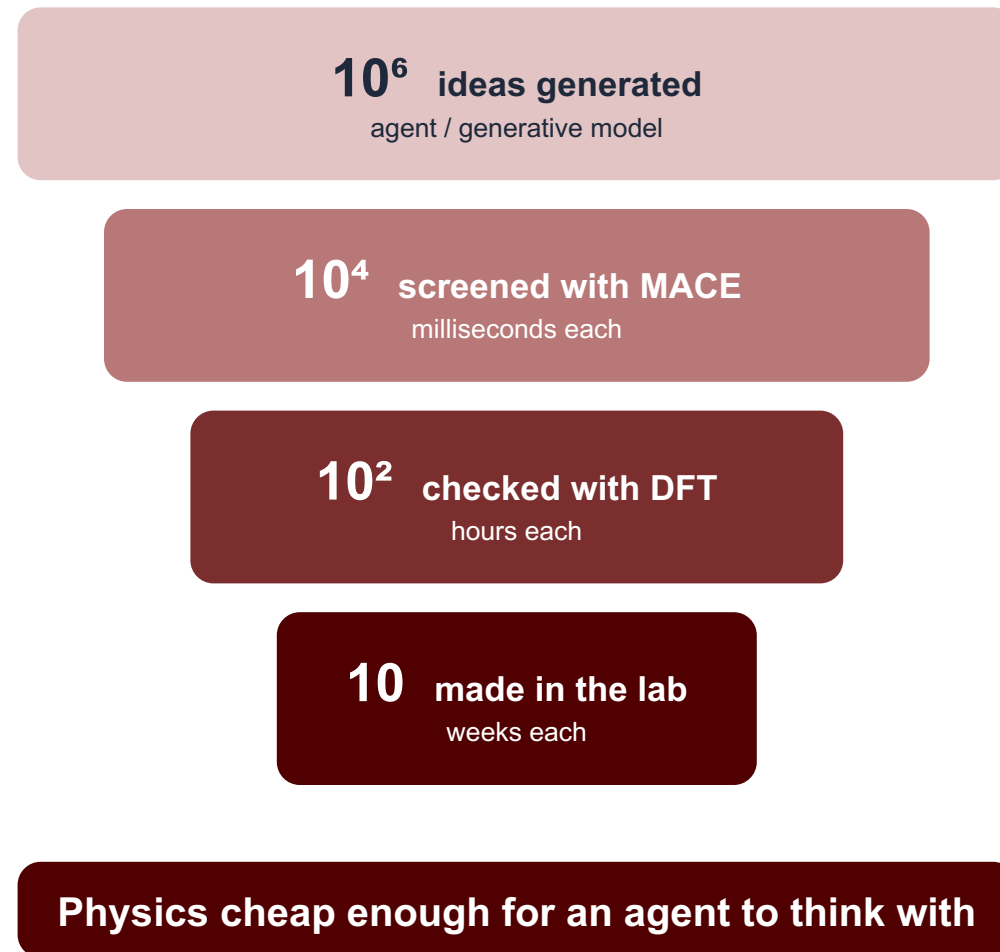
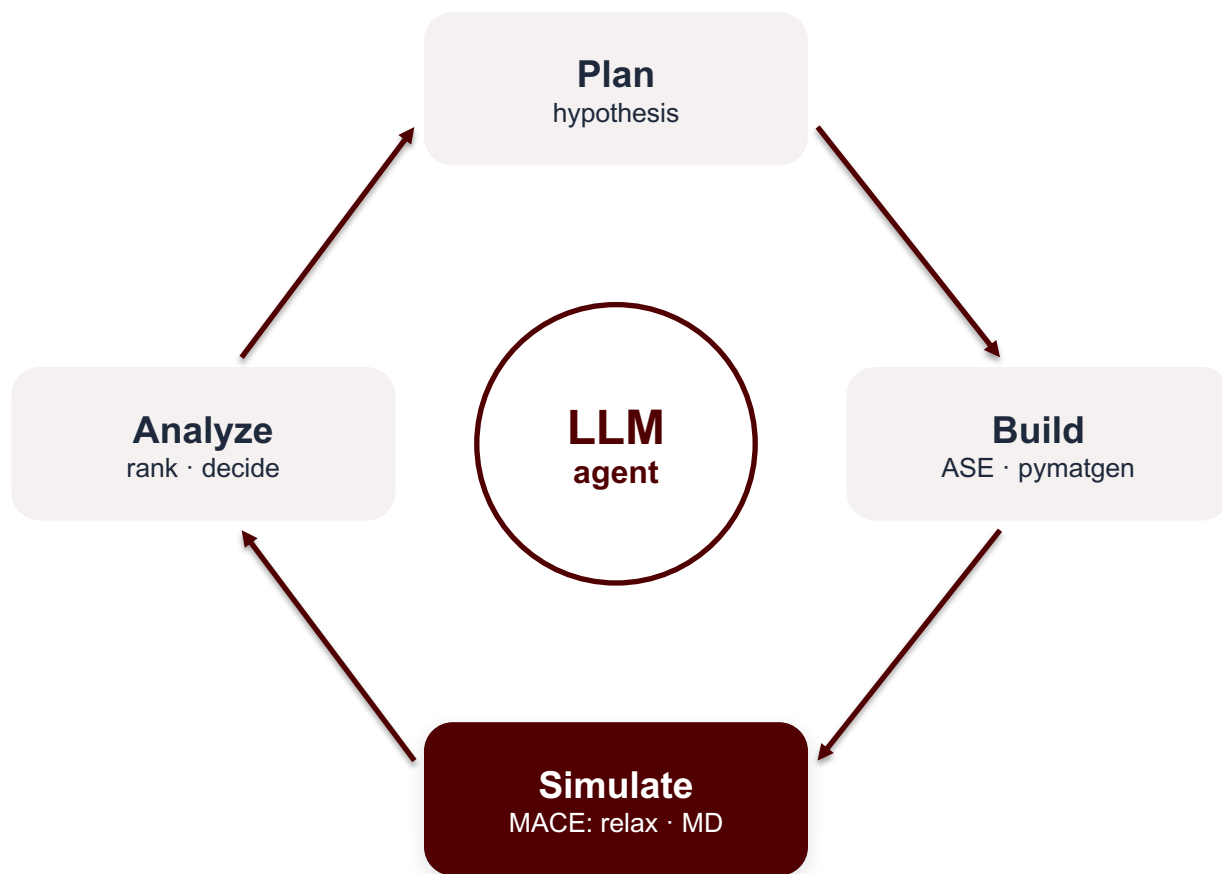
Inside MACE: many-body equivariant messages

A graph neural network where atoms are nodes and neighbors within r_c are edges



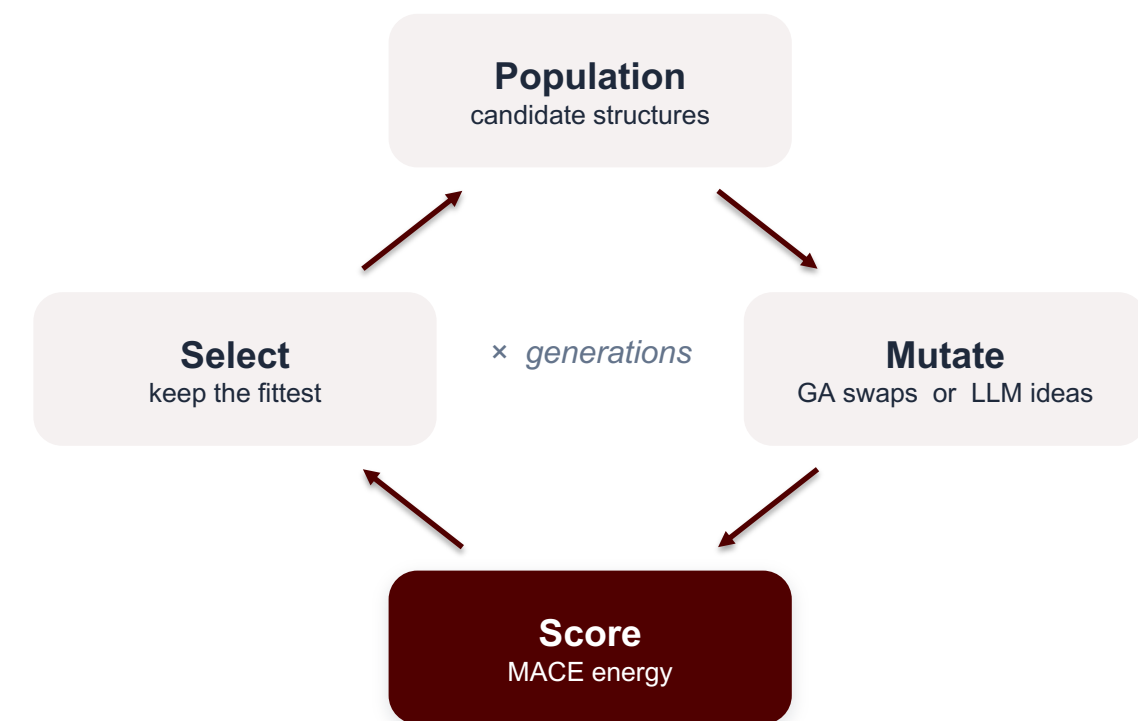
MACE as the agent's physics tool

The LLM plans and reasons; MACE answers “what would the atoms do?” in milliseconds



Evolution with MACE as the fitness function

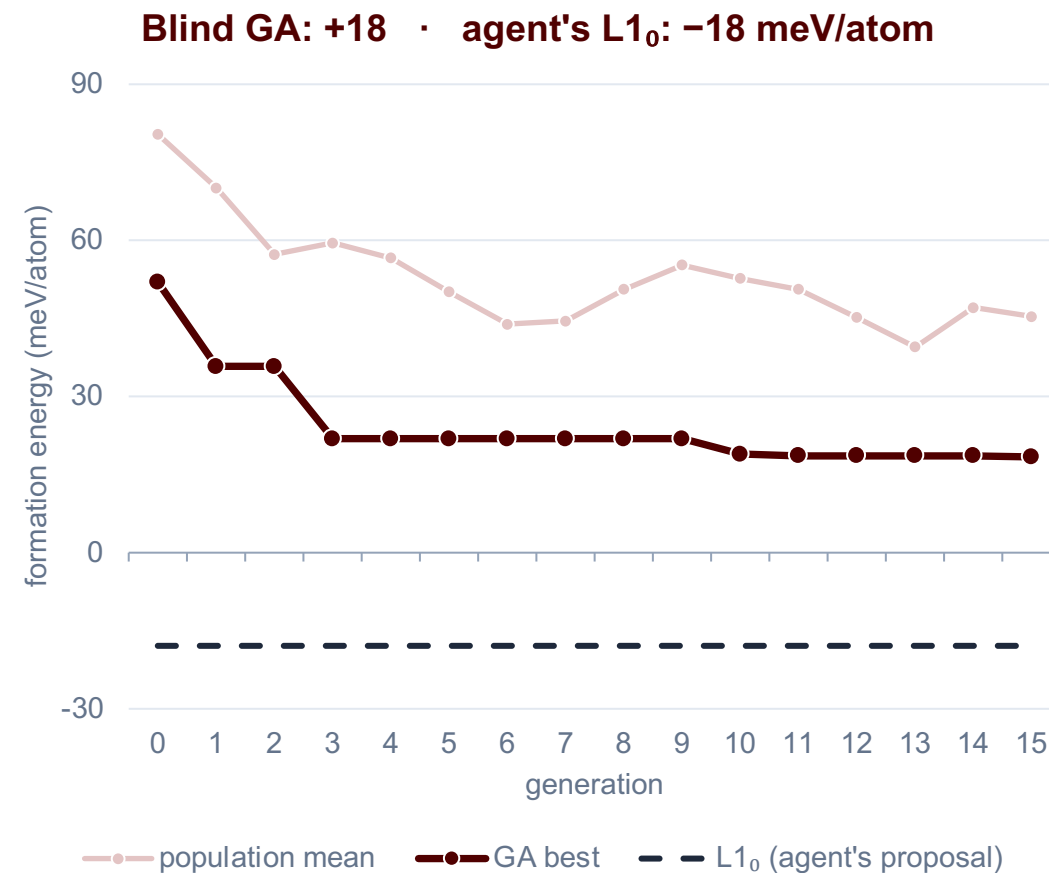
Generate, score, select, repeat: the LLM can be the mutation operator



GENOME: 32 SITES, 4 LAYERS (shown: L1₀ CuAu, the known stable phase)



■ Cu ■ Au



More ways to pair MACE with an agent

Same tool, four workflow shapes

1 Active learning

MACE unsure?

DFT labels it

fine-tune MACE

the agent decides where the model needs more data

2 MCP tool server

Claude

relax · md · phonons

MACE

expose MACE as tools any agent can call

3 Multi-agent team

Planner

Simulator (MACE)

Critic

one proposes, one computes, one checks

4 Property prediction

Agent sets T, P

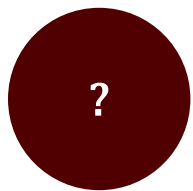
MACE dynamics

diffusion, strength

long simulations that DFT can't afford

Trust, but verify

Pressure-test MACE before an agent acts on it



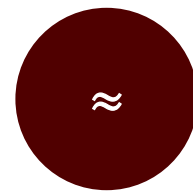
Out of distribution

Chemistry far from the training data fails silently. Watch uncertainty.



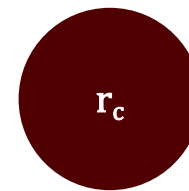
Inherits DFT errors

Trained on PBE-level DFT, so its known biases carry over.



Softened surface

Universal models tend to under-predict stiffness. Fine-tune for production.



Local by design

A ~ 6 Å cutoff misses long-range electrostatics without add-ons.

Agent rule: MACE proposes, DFT disposes. Spot-check every winner.

LIVE DEMO

Claude Code × MACE

An LLM agent in VS Code using MACE as its physics tool, live on this laptop

1

Forces

Stretch a water molecule, read the forces, watch it relax

2

Blind evolution

A genetic algorithm scores 235 Cu/Au alloys in 17 s

3

Agent evolution

Claude proposes structures, MACE judges them

Key takeaways

Four ideas to leave with

$E \rightarrow F$

1

One model, all forces

Learn the energy; forces and stress come free by differentiation.

v

2

Physics built in

Equivariant, many-body messages make it accurate and fast.

AI

3

Agents need cheap physics

MACE turns an LLM's hypotheses into numbers in milliseconds.

✓

4

Close the loop

Evolve with MACE, verify winners with DFT and the lab.

Questions?